

=> Foundation of Bayes Theorem:

$$P(y | x_1, x_2, \dots, x_n) = \frac{P(x_1, x_2, \dots, x_n | y) \times P(y)}{P(x_1, x_2, \dots, x_n)}$$

↑
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Posterior Probability
Evidence

Likelihood Prior

=> Formulating the Predictor:

$$P(y | x_1, \dots, x_n) = P(y) \prod P(x_i | y)$$

$$\hat{y} = \operatorname{argmax} [P(y) \prod P(x_i | y)]$$

=> Solving an example: Find the probability of playing tennis if the outlook is given as sunny, the temp is cool, the humidity is high &

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes

is given as sunny, the temp is cool, the humidity is high &

Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

humidity is high Δ
wind is true.

$$P(\text{Play Tennis}) = 9/14 \quad \left. \begin{array}{l} \\ \end{array} \right\} P(y)$$

$$P(\text{Play Tennis} (n)) = 5/14 \quad \left. \begin{array}{l} \\ \end{array} \right\} P(y)$$

Outlook	y	n
Sunny	2/9	3/5
Overcast	4/9	0
Rainy	3/9	2/5

Temp	y	n
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	y	n
High	3/9	4/5
Normal	6/9	1/5

Windy	y	n
True	3/9	3/5
False	6/9	2/5

$$P(\text{Play Tennis} | \text{yes}) = P(y) \times P(\text{sunny} | y) \times P(\text{cool} | y) \times P(\text{high} | y) \times P(\text{true} | y)$$

$$= \frac{9}{14} \times \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} = \boxed{0.005}$$

$$\begin{aligned}
 P(\text{Play Tennis}(N_o)) &= P(N_o) \times P(\text{Sunny}(N_o)) \times P(\text{Cool}(N_o)) \times \\
 &\quad P(\text{High}(N_o)) \times P(\text{True}(N_o)) \\
 &= \frac{5}{14} \times \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} = \boxed{0.0206}
 \end{aligned}$$

Verdict: When it is sunny, & cool with humidity $\approx$ high & wind is true we do not play tennis.

$$\begin{aligned}
 P(y) &= \frac{P(y)}{P(N) + P(y)} & \text{Normalization} & \quad P(N) = \frac{P(N)}{P(y) + P(N)} \\
 &= \frac{0.005}{0.0206 + 0.005} & \text{Probability} & \quad = \frac{0.0206}{0.005 + 0.0206} \\
 &= 0.1753 & & \quad = 0.8046 \checkmark
 \end{aligned}$$

$\Rightarrow$ Non-Parametric Models:

$\hookrightarrow$ No predetermined parameters are possible.

1. Parametric - (Recall & Risk)

↳ Requires optimism - (because of bias)